CATMS-backend

Database Authorship Report

Viva / Oral Defence Reference

Investigation type	Read-only · git log --all + git blame
Branches covered	main, develop, malini, nadish-backend, thiseni, member5-billing-reports
Files audited	20 database files (sql/, database/triggers, procedures, functions, views)
Authors identified	5 · Nadish Sathsara · aswieni · Malini Thisaruka · Sehath Thewjith · Thiseni Abeykoon
Report date	2026-10-13

Prepared for: Oral Viva / Defence | **Date:** 2026-10-13 | **Read-only — no files changed**

Step 4 — GitHub Username ↔ Real Name Mapping

(Source: git commit email addresses + `TEAM_WORKLOAD_DISTRIBUTION.md`)

Git Name (commit)	Git Email	Member #	Domain Responsibility
Nadish Sathsara (`nadish-backend` branch)	nadishsathsara2003@gmail.com	**Member 1 — Coordinator**	Auth, Admin, Branches, Core Foundation, Schema owner
aswieni / Aswieni Udhayakumar (`malini` / `develop` branch reviewer commits)	aswieniu2004@gmail.com	**Member 2**	Patients & Insurance (also acted as schema constraint reviewer/fixer)
Malini Thisaruka / mthisaruka (`malini` branch)	mthisaruka@gmail.com	**Member 3**	Doctors, Appointments, Scheduling
Nadish Sathsara (on `nadish-backend`)	—	Also **Member 4 domain**	Consultation→Invoice sp (written by M4 scope but committed by M1/coordinator)
Sehath Thewjith / Sehath50 (`member5-billing-reports` branch)	sehathsilva@gmail.com	**Member 5**	Billing, Payments, Invoice functions
Thiseni Abeykoon (`thiseni` branch)	thiseniabeykoon.24@cse.mrt.ac.lk	**Member 1 support / Admin**	Admin procedures, Staff trigger, views

`Thiseni Abeykoon` *committed Admin/Branch procedures (sp_force_password_reset, sp_transfer_branch_manager) and views — this aligns with Member 1's (Admin & Branches) domain scope as documented in TEAM_WORKLOAD_DISTRIBUTION.md. Thiseni is a secondary contributor to Member 1's area.*

Step 1 — Complete Database File Inventory

`sql/` — Core SQL scripts

File	Description
[`sql/01_schema.sql`](file:///c:/Users/HP/Downloads/CATMS-System%20%2011/CATMS-System/CATMS-backend/sql/01_schema.sql)	Full 19-table DDL schema (all CREATE TABLE statements)
[`sql/02_seed_data.sql`](file:///c:/Users/HP/Downloads/CATMS-System%20%2011/CATMS-System/CATMS-backend/sql/02_seed_data.sql)	Initial seed data (staff, branches, patients, doctors)
[`sql/03_indexes.sql`](file:///c:/Users/HP/Downloads/CATMS-System%20%2011/CATMS-System/CATMS-backend/sql/03_indexes.sql)	Performance indexes
[`sql/run_all.py`](file:///c:/Users/HP/Downloads/CATMS-System%20%2011/CATMS-System/CATMS-backend/sql/run_all.py)	Python script: loads schema + seed + triggers/procedures/views
[`sql/init_if_empty.py`](file:///c:/Users/HP/Downloads/CATMS-System%20%2011/CATMS-System/CATMS-backend/sql/init_if_empty.py)	Docker first-start init guard
[`sql/seed_demo_data.py`](file:///c:/Users/HP/Downloads/CATMS-System%20%2011/CATMS-System/CATMS-backend/sql/seed_demo_data.py)	Extended demo data for marking (extra marks submission)

`database/triggers/`

File	Description
[`trg_check_appointment_overlap.sql`](file:///c:/Users/HP/Downloads/CATMS-System%20%2011/CATMS-System/CATMS-backend/database/triggers/trg_check_appointment_overlap.sql)	Prevents double-booking same doctor same timeslot

File	Description
[trg_audit_core_entities.sql](file:///c:/Users/HP/Downloads/CATMS-System%20%2011/CATMS-System/CATMS-backend/database/triggers/trg_audit_core_entities.sql)	Writes to Audit_Log on INSERT/UPDATE across 8 tables
[trg_after_staff_update.sql](file:///c:/Users/HP/Downloads/CATMS-System%20%2011/CATMS-System/CATMS-backend/database/triggers/trg_after_staff_update.sql)	Audits staff record changes
[trg_insurance_claim_rules.sql](file:///c:/Users/HP/Downloads/CATMS-System%20%2011/CATMS-System/CATMS-backend/database/triggers/trg_insurance_claim_rules.sql)	Enforces insurance claim amount limits and rules

database/procedures/

File	Description
[sp_book_appointment.sql](file:///c:/Users/HP/Downloads/CATMS-System%20%2011/CATMS-System/CATMS-backend/database/procedures/sp_book_appointment.sql)	Atomic appointment slot allocation stored procedure
[sp_complete_consultation_and_generate_invoice.sql](file:///c:/Users/HP/Downloads/CATMS-System%20%2011/CATMS-System/CATMS-backend/database/procedures/sp_complete_consultation_and_generate_invoice.sql)	Completes consultation + auto-generates invoice atomically
[sp_force_password_reset.sql](file:///c:/Users/HP/Downloads/CATMS-System%20%2011/CATMS-System/CATMS-backend/database/procedures/sp_force_password_reset.sql)	Admin procedure to force user password reset
[sp_record_payment.sql](file:///c:/Users/HP/Downloads/CATMS-System%20%2011/CATMS-System/CATMS-backend/database/procedures/sp_record_payment.sql)	Records payment and updates Invoice status
[sp_transfer_branch_manager.sql](file:///c:/Users/HP/Downloads/CATMS-System%20%2011/CATMS-System/CATMS-backend/database/procedures/sp_transfer_branch_manager.sql)	Transfers manager between branches atomically

database/functions/

File	Description
[fn_calculate_patient_balance.sql](file:///c:/Users/HP/Downloads/CATMS-System%20%2011/CATMS-System/CATMS-backend/database/functions/fn_calculate_patient_balance.sql)	Calculates outstanding patient balance (invoice - insurance - payments)
[fn_policy_coverage.sql](file:///c:/Users/HP/Downloads/CATMS-System%20%2011/CATMS-System/CATMS-backend/database/functions/fn_policy_coverage.sql)	Returns coverage % for a patient+treatment combination

database/views/

File	Description
[vw_staff_directory.sql](file:///c:/Users/HP/Downloads/CATMS-System%20%2011/CATMS-System/CATMS-backend/database/views/vw_staff_directory.sql)	Joins Staff + User_Account + Branch for admin UI
[vw_Invoice_Summary.sql](file:///c:/Users/HP/Downloads/CATMS-System%20%2011/CATMS-System/CATMS-backend/database/views/vw_Invoice_Summary.sql)	Invoice with treatment fee totals
[vw_audit_log_readable.sql](file:///c:/Users/HP/Downloads/CATMS-System%20%2011/CATMS-System/CATMS-backend/database/views/vw_audit_log_readable.sql)	Human-readable audit log joining usernames and table names

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Step 2+3 — Full Authorship History & Attribution Table

`sql/01_schema.sql` — 27 commits, 2 primary authors

Commit	Date	Author	What Changed
`8caada1`	2026-09-30	**Nadish Sathsara**	**Initial schema** — all 19 tables, all FKs, all ENUMs, DROP sequence
`64b7f50`	2026-10-01	**aswieni**	Removed erroneous `UNIQUE` on `Patient.Contact_Number`
`b3f204d`	2026-10-01	**aswieni**	Added `UNIQUE` on `Branch.Branch_Name`
`c0718bb`	2026-10-01	**aswieni**	Added `UNIQUE` on `Insurance_Provider.Provider_Name`
`07222c1`	2026-10-01	**aswieni**	Changed `Policy_Number` → composite UNIQUE(Provider_ID, Policy_Number)
`46b261c`	2026-10-01	**aswieni**	Added index `Appointment(Doctor_ID, Appointment_Date)` for overlap perf
`13ebc55`	2026-10-01	**aswieni**	Added `UNIQUE` on `Staff.Email`
`37f1a0e`	2026-10-01	**aswieni**	Added CHECK constraints on `Insurance_Claim` (amounts, settlement date)
`629fc12`	2026-10-01	**aswieni**	Added CHECK on `Appointment.Duration_Minutes`
`787b445`	2026-10-01	**aswieni**	Added CHECK on `Doctor.Standard_Consultation_Fee`
`3453213`	2026-10-01	**aswieni**	Added CHECK on `Treatment_Catalogue.Standard_Unit_Price`
`463cc84`	2026-10-01	**aswieni**	Added CHECK on `Prescribed_Treatment.Billed_Unit_Price`
`49a3154`	2026-10-01	**aswieni**	Added CHECK on `Invoice.Billed_Consultation_Fee`
`16c179a`	2026-10-01	**aswieni**	Added CHECK on `Treatment_Policy_Eligibility.Covered_Percentage`
`a359b96`	2026-10-01	**aswieni**	Added CHECK on `Insurance_Policy.Default_Coverage_Percentage`
`a6c5b76`	2026-10-01	**aswieni**	Added composite UNIQUE on `Doctor_Schedule` slots (prevent duplicate slots)
`6d41a57`	2026-10-01	**aswieni**	Added CHECK `Consultation.Follow_Up_Date` must not be before consult date
`4b66b85`	2026-10-01	**aswieni**	Added NOT NULL + mandatory cancellation reason on `Appointment`
`f2b1e46`	2026-10-01	**aswieni**	Added `Patient.Updated_At` column (optimistic concurrency)
`152afb5`	2026-10-06	**aswieni**	Fixed `Patient.Updated_At` precision to `DATETIME(6)`
`f8946e4`	2026-10-06	**aswieni**	(duplicate fix attempt, same issue)
`21596c5`	2026-10-09	**Malini Thisaruka**	Added `In_Progress` value to Appointment.Status ENUM

Commit	Date	Author	What Changed
`58620d1`	2026-10-10	**Aswieni Udhayakumar**	Added `Patient_Portal_Invite` table
`33590b0`	2026-10-11	**aswieni**	Removed `Patient_Portal_Invite` table (reverted)

Blame summary for 01_schema.sql:

- ~95% of lines: **Nadish Sathsara** (entire original schema structure, all tables, all FKs)
- Key individual lines modified by **aswieni**: `Branch.Branch_Name UNIQUE` (line 78), `Staff.Email UNIQUE` (line 85), `Insurance_Provider.Provider_Name UNIQUE` (line 311), `Patient.Contact_Number` (line 267, removed UNIQUE), `Patient.Updated_At` (line 278), `Doctor_Schedule` composite UNIQUE (lines 238-241)
- 1 line modified by **Malini Thisaruka**: `Appointment.Status` ENUM addition of `In_Progress` (line 24 of trigger, and the schema ENUM value)

`sql/02_seed_data.sql` — 5 commits, 3 authors

Commit	Date	Author	What Changed
`8caada1`	2026-09-30	**Nadish Sathsara**	All original seed rows: 3 branches, 9 staff, all doctors, patients, schedules, appointments, consultations
`38d81e4`	2026-09-30	**Nadish Sathsara**	Switched to local Docker MySQL
`aab4643`	2026-10-10	**Malini Thisaruka**	Added valid bcrypt hashes for all `Password123!` seed accounts
`96b2111`	2026-10-10	**Malini Thisaruka**	Final verified bcrypt hash values (`$2b$12$W.5CSw...`)
`7cdbdcc`	2026-10-10	**Nadish Sathsara**	Added accounts 14–16 (`pat_ananya`, `billing_kandy`, `billing_galle`) + Staff 10–12 for branch-specific billing

Blame summary:

- Lines 1–52 (structure + users 1–13, branches 1–3, staff 1–9): **Nadish Sathsara** (original structure) with **Malini Thisaruka** overwriting the `Password_Hash` column values (lines 40–52, hash values rewritten Oct 10)
- Lines 53–55 (accounts 14–16) and lines 78–80 (staff 10–12): **Nadish Sathsara**
- **The rest of the file** (Doctor inserts, Patient inserts, Appointment inserts, Consultation inserts, Invoice/Payment inserts): **Nadish Sathsara**

`sql/03_indexes.sql` — 1 commit, 1 author

Commit	Date	Author	What
`979a287`	2026-10-12	**Nadish Sathsara**	Entire file — performance indexes (extra marks submission)

100% Nadish Sathsara. Written as part of the "Fix 15: Indexes, demo data and CI" extra-marks submission.

`sql/run_all.py` — 9 commits, 4 authors

Commit	Date	Author	What Changed
`c5f913d`	2026-09-30	**Nadish Sathsara**	Initial script created
`812f886`	2026-09-30	**Nadish Sathsara**	Made SSL optional for local MySQL
`38d81e4`	2026-09-30	**Nadish Sathsara**	Docker MySQL switch

Commit	Date	Author	What Changed
`551d62e`	2026-10-08	**Sehath Thewjith**	Billing integration fixes
`19ad8fd`	2026-10-09	**Malini Thisaruka**	Fixed endpoint exports, router imports, db routine runner
`3f918bc`	2026-10-09	**Malini Thisaruka**	Load all routines and triggers by default
`4eccbdc`	2026-10-10	**Malini Thisaruka**	Audit logging integration
`5d676f7`	2026-10-12	**Nadish Sathsara**	Fix 4: Insurance claim limits
`0db6f91`	2026-10-12	**Nadish Sathsara**	CA certificate support for cloud SSL

Primary authors: Nadish Sathsara (created + cloud/SSL fixes) and **Malini Thisaruka** (routine loading logic, trigger wiring).

`sql/init_if_empty.py` — 1 commit, 1 author

Commit	Date	Author	What
`f9bdc28`	2026-10-12	**Nadish Sathsara**	Entire file — Docker first-start guard (Fix 8)

100% Nadish Sathsara.

`sql/seed_demo_data.py` — 1 commit, 1 author

Commit	Date	Author	What
`992ade7`	2026-10-12	**Nadish Sathsara**	Entire file — extended demo data for extra marks (Fix 15)

100% Nadish Sathsara.

`database/triggers/trg_check_appointment_overlap.sql` — 5 commits, 2 authors

Commit	Date	Author	What Changed
`151eaee`	2026-09-26	**Malini Thisaruka**	Initial creation of the BEFORE INSERT overlap trigger
`04bfebf`	2026-09-27	**Malini Thisaruka**	Refactored to isolate common base
`55d4b8b`	2026-09-27	**Malini Thisaruka**	Added BEFORE UPDATE trigger (covers rescheduling); all 88 lines
`3f918bc`	2026-10-09	**Malini Thisaruka**	Added `FOR UPDATE` row lock (concurrency fix) — lines 18, 25–26, 59, 66–67
`21596c5`	2026-10-09	**Malini Thisaruka**	Added `In_Progress` to status check ENUMs — lines 24, 33, 65, 75

100% Malini Thisaruka (all 88 lines, all 5 commits). The BEFORE UPDATE trigger for rescheduling was her addition — not just the insert trigger.

`database/triggers/trg_audit_core_entities.sql` — 1 commit, 1 author

Commit	Date	Author	What
`4eccbdc`	2026-10-10	**Malini Thisaruka**	Entire file — all 460 lines

100% Malini Thisaruka. This is the largest trigger file — covers AFTER INSERT/UPDATE triggers on: `Patient`, `Staff`, `Appointment`, `Consultation`, `Invoice`, `Payment`, `Insurance_Claim` — 8 entities, 14 trigger definitions total.

`database/triggers/trg_after_staff_update.sql` — 1 commit, 1 author

Commit	Date	Author	What
`a7598e9`	2026-10-08	**Thiseni Abeykoon**	Entire file

100% Thiseni Abeykoon (Member 1 / Admin domain support).

`database/triggers/trg_insurance_claim_rules.sql` — 1 commit, 1 author

Commit	Date	Author	What
`e8d1df2`	2026-10-12	**Nadish Sathsara**	Entire file (Fix 4: Insurance claim limits and invoice status)

100% Nadish Sathsara.

`database/procedures/sp_book_appointment.sql` — 1 commit, 1 author

Commit	Date	Author	What
`6b2d107`	2026-09-28	**Malini Thisaruka**	Entire file — atomic appointment booking SP

100% Malini Thisaruka.

`database/procedures/sp_complete_consultation_and_generate_invoice.sql` — 1 commit, 1 author

Commit	Date	Author	What
`caadd1f`	2026-10-04	**Nadish Sathsara**	Entire file — generates invoice when consultation completes

100% Nadish Sathsara. Note: this procedure implements **Member 4's domain** (consultation→invoice) but was committed by the coordinator (Member 1/Nadish) during integration review.

`database/procedures/sp_force_password_reset.sql` — 1 commit, 1 author

Commit	Date	Author	What
`a7598e9`	2026-10-08	**Thiseni Abeykoon**	Entire file

100% Thiseni Abeykoon (Admin/Security domain, Member 1 area).

`database/procedures/sp_record_payment.sql` — 2 commits, 1 author

Commit	Date	Author	What Changed
`082ca73`	2026-10-04	**Sehath Thewjith**	Initial SP creation
`551d62e`	2026-10-08	**Sehath Thewjith**	Billing integration/validation fixes

100% Sehath Thewjith (Member 5).

`database/procedures/sp_transfer_branch_manager.sql` — 1 commit, 1 author

Commit	Date	Author	What
`a7598e9`	2026-10-08	**Thiseni Abeykoon**	Entire file

100% Thiseni Abeykoon (Branch Management domain, Member 1 area).

`database/functions/fn_calculate_patient_balance.sql` — 1 commit, 1 author

Commit	Date	Author	What
`082ca73`	2026-10-04	**Sehath Thewjith**	Entire file

100% Sehath Thewjith (Member 5).

`database/functions/fn_policy_coverage.sql` — 1 commit, 1 author

Commit	Date	Author	What
`08bfb3b`	2026-10-12	**Nadish Sathsara**	Entire file (Fix 4: Insurance claim limits)

100% Nadish Sathsara.

`database/views/vw_staff_directory.sql` — 1 commit, 1 author

Commit	Date	Author	What
`a7598e9`	2026-10-08	**Thiseni Abeykoon**	Entire file

100% Thiseni Abeykoon.

`database/views/vw_Invoice_Summary.sql` — 2 commits, 2 authors

Commit	Date	Author	What
`082ca73`	2026-10-04	**Sehath Thewjith**	Initial view creation
`58a8b57`	2026-10-11	**Nadish Sathsara**	Extended to include treatment fees in invoice totals

Shared ownership: Original view by **Sehath Thewjith**; extended by **Nadish Sathsara** to fix invoice total calculations.

`database/views/vw_audit_log_readable.sql` — 1 commit, 1 author

Commit	Date	Author	What
`a7598e9`	2026-10-08	**Thiseni Abeykoon**	Entire file

100% Thiseni Abeykoon.

Step 3 — Master Attribution Table (Sorted by File)

File	Original Author (1st commit)	All Contributors	Most Recent Author	# Commits	Specific Contributions
`sql/01_schema.sql`	**Nadish Sathsara** `8caada1` (2026-09-30)	Nadish + aswieni + Malini	aswieni `33590b0` (2026-10-11)	**27**	Nadish: full 19-table DDL, all FKs, ENUMs; aswieni: 15+ commits of CHECK constraints, UNIQUE constraints, `Patient.Updated_At`, schema corrections; Malini: `In_Progress` ENUM value `21596c5`
`sql/02_seed_data.sql`	**Nadish Sathsara** `8caada1` (2026-09-30)	Nadish + Malini	Nadish `7cdbdcc` (2026-10-10)	**5**	Nadish: all entity data rows (branches, staff, doctors, patients, appointments, invoices); Malini: rewrote all `Password_Hash` values to valid bcrypt `96b2111`
`sql/03_indexes.sql`	**Nadish Sathsara** `979a287` (2026-10-12)	Nadish only	Nadish	**1**	100% Nadish — extra marks indexes
`sql/run_all.py`	**Nadish Sathsara** `c5f913d` (2026-09-30)	Nadish + Malini + Sehath	Nadish `0db6f91` (2026-10-12)	**9**	Nadish: initial script + SSL/cloud; Malini: routine/trigger loader logic; Sehath: billing fix
`sql/init_if_empty.py`	**Nadish Sathsara** `f9bdc28` (2026-10-12)	Nadish only	Nadish	**1**	100% Nadish
`sql/seed_demo_data.py`	**Nadish Sathsara** `992ade7` (2026-10-12)	Nadish only	Nadish	**1**	100% Nadish
`database/triggers/trg_check_appointment_overlap.sql`	**Malini Thisaruka** `151eaee` (2026-09-26)	Malini only	Malini `21596c5` (2026-10-09)	**5**	100% Malini — BEFORE INSERT + BEFORE UPDATE overlap triggers, concurrency row locks
`database/triggers/trg_audit_core_entities.sql`	**Malini Thisaruka** `4eccbdc` (2026-10-10)	Malini only	Malini	**1**	100% Malini — 460-line file, 14 triggers across 8 tables
`database/triggers/trg_after_staff_update.sql`	**Thiseni Abeykoon** `a7598e9` (2026-10-08)	Thiseni only	Thiseni	**1**	100% Thiseni
`database/triggers/trg_insurance_claim_rules.sql`	**Nadish Sathsara** `e8d1df2` (2026-10-12)	Nadish only	Nadish	**1**	100% Nadish
`database/procedures/sp_book_appointment.sql`	**Malini Thisaruka** `6b2d107` (2026-09-28)	Malini only	Malini	**1**	100% Malini
`database/procedures/sp_complete_consultation_and_generate_invoice.sql`	**Nadish Sathsara** `caadd1f` (2026-10-04)	Nadish only	Nadish	**1**	100% Nadish (coordinator implementing M4 scope)
`database/procedures/sp_force_password_reset.sql`	**Thiseni Abeykoon** `a7598e9` (2026-10-08)	Thiseni only	Thiseni	**1**	100% Thiseni

File	Original Author (1st commit)	All Contributors	Most Recent Author	# Commits	Specific Contributions
database/procedures/sp_record_payment.sql	**Sehath Thewjith** 082ca73 (2026-10-04)	Sehath only	Sehath 551d62e (2026-10-08)	**2**	100% Sehath
database/procedures/sp_transfer_branch_manager.sql	**Thiseni Abeykoon** a7598e9 (2026-10-08)	Thiseni only	Thiseni	**1**	100% Thiseni
database/functions/fn_calculate_patient_balance.sql	**Sehath Thewjith** 082ca73 (2026-10-04)	Sehath only	Sehath	**1**	100% Sehath
database/functions/fn_policy_coverage.sql	**Nadish Sathsara** 08bfb3b (2026-10-12)	Nadish only	Nadish	**1**	100% Nadish
database/views/vw_staff_directory.sql	**Thiseni Abeykoon** a7598e9 (2026-10-08)	Thiseni only	Thiseni	**1**	100% Thiseni
database/views/vw_Invoice_Summary.sql	**Sehath Thewjith** 082ca73 (2026-10-04)	Sehath + Nadish	Nadish 58a8b57 (2026-10-11)	**2**	Sehath: initial view; Nadish: extended to include treatment fee totals
database/views/vw_audit_log_readable.sql	**Thiseni Abeykoon** a7598e9 (2026-10-08)	Thiseni only	Thiseni	**1**	100% Thiseni

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Step 5 — Viva-Ready Summary (Oral Defence Cheat Sheet)

Q1: Who wrote the original schema (`01_schema.sql`) and what was their initial design?

Nadish Sathsara created the entire initial schema in commit `8caada1` (2026-09-30). His design covers all 19 tables organized across 5 domains:

- **Security/Auth**: `User_Account`, `Audit_Log`
- **Organization**: `Branch`, `Staff`, `Doctor`, `Specialty`, `Doctor_Specialty`, `Doctor_Schedule`
- **Patients/Insurance**: `Patient`, `Emergency_Contact`, `Insurance_Provider`, `Insurance_Policy`
- **Appointments**: `Appointment`
- **Clinical/Billing**: `Treatment_Category`, `Treatment_Catalogue`, `Treatment_Policy_Eligibility`, `Consultation`, `Prescribed_Treatment`, `Invoice`, `Insurance_Claim`, `Payment`

After the initial commit, **aswieni (Member 2)** contributed **15 separate fix/constraint commits** on 2026-10-01 that hardened the schema: adding 8 CHECK constraints (prices, durations, amounts), 4 UNIQUE constraints (Staff.Email, Branch.Branch_Name, Provider_Name, composite Policy_Number), the `Patient.Updated_At` optimistic concurrency column, and the composite UNIQUE on Doctor_Schedule slots.

Malini Thisaruka added `In_Progress` to the `Appointment.Status` ENUM in commit `21596c5` (2026-10-09).

Q2: Who added which triggers and what does each do?

Trigger	Author	Commit	What It Does
trg_check_appointment_overlap_insert	**Malini Thisaruka**	55d4b8b (2026-09-27)	BEFORE INSERT on Appointment — rejects new booking if doctor has overlapping active appointment
trg_check_appointment_overlap_update	**Malini Thisaruka**	55d4b8b (2026-09-27)	BEFORE UPDATE on Appointment — same check for reschedules

Trigger	Author	Commit	What It Does
(concurrency lock added to both)	**Malini Thisaruka**	`3f918bc` (2026-10-09)	`SELECT ... FOR UPDATE` to serialize concurrent bookings for same doctor
`trg_after_staff_update`	**Thiseni Abeykoon**	`a7598e9` (2026-10-08)	AFTER UPDATE on Staff — logs staff changes to Audit_Log
`trg_audit_core_entities` (14 triggers)	**Malini Thisaruka**	`4eccbdc` (2026-10-10)	AFTER INSERT/UPDATE on 8 tables (Patient, Staff, Appointment, Consultation, Invoice, Payment, Insurance_Claim, etc.) — persistent audit trail using MySQL session variables
`trg_insurance_claim_rules`	**Nadish Sathsara**	`e8d1df2` (2026-10-12)	Enforces claim amount cannot exceed invoice total; validates approved ≤ claimed

Q3: Who wrote which stored procedures/functions and what do they do?

Object	Author	Commit	What It Does
`sp_book_appointment`	**Malini Thisaruka**	`6b2d107` (2026-09-28)	Atomically inserts appointment slot with validation — prevents race conditions
`sp_complete_consultation_and_generate_invoice`	**Nadish Sathsara**	`caadd1f` (2026-10-04)	Completes a consultation, records prescribed treatments, auto-generates the Invoice in one transaction
`sp_force_password_reset`	**Thiseni Abeykoon**	`a7598e9` (2026-10-08)	Admin sets `must_reset = TRUE` on any user account
`sp_transfer_branch_manager`	**Thiseni Abeykoon**	`a7598e9` (2026-10-08)	Atomically reassigns Branch.Manager_Staff_ID to new staff member
`sp_record_payment`	**Sehath Thewjith**	`082ca73` (2026-10-04) + `551d62e` (2026-10-08)	Records payment against invoice; auto-updates Invoice status to `Partially_Paid` or `Paid`
`fn_calculate_patient_balance`	**Sehath Thewjith**	`082ca73` (2026-10-04)	SQL function: returns (invoice total – insurance approved – payments made)
`fn_policy_coverage`	**Nadish Sathsara**	`08bfb3b` (2026-10-12)	Returns coverage % for a given patient+policy+treatment combination

Q4: Who wrote the seed data and demo data scripts?

- `sql/02_seed_data.sql`: **Nadish Sathsara** wrote all original rows (branches, staff, doctors, patients, appointments, consultations, invoices). **Malini Thisaruka** rewrote the `Password_Hash` column values for all accounts to valid bcrypt hashes in commit `96b2111` (2026-10-10) after discovering the original placeholders were invalid.
- `sql/seed_demo_data.py`: **100% Nadish Sathsara** — `992ade7` (2026-10-12), extended demo dataset for extra marks.

Q5: Who set up indexes?

100% Nadish Sathsara — `sql/03_indexes.sql`, commit `979a287` (2026-10-12) as part of "Fix 15: Indexes, demo data and CI (extra marks)". This file was created as an extra-marks submission and contains performance indexes for common query patterns (e.g. appointment date+doctor lookups). An earlier inline index on `Appointment(Doctor_ID, Appointment_Date)` was added by **aswieni** directly in `01_schema.sql` commit `46b261c` (2026-10-01) for the overlap trigger's performance.

Q6: Files with shared/unclear ownership

File	Situation
`sql/01_schema.sql`	■■ **Shared** — Nadish wrote the structure; aswieni added 15 constraint commits. Approximately 90% of current lines are Nadish's original code; the constraint lines (UNIQUE, CHECK) are aswieni's. Both contributed significantly but with different *types* of work.
`sql/02_seed_data.sql`	■■ **Shared** — Nadish wrote data rows; Malini rewrote all `Password_Hash` values (all accounts, lines 40–55 by blame). Data ownership = Nadish; hash/auth fix = Malini.
`sql/run_all.py`	■■ **Shared** — 4 authors touched this: Nadish (created, SSL, cloud), Malini (routine loader), Sehath (billing fix), Nadish again (final fixes).
`database/views/vw_Invoice_Summary.sql`	■■ **Shared** — Sehath created it; Nadish extended it to include treatment fee totals.

All data above is derived from `git log --all --follow` *and* `git blame`*. These are verifiable on GitHub — each commit hash can be checked at* `https://github.com/CATMS-System/CATMS-backend/commit/`*.*